

COURSE TERM PROJECT

Final report

Team:水色真珠音調~用閃亮的歌聲開始現場演唱

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AI council finetuning

Parameter

Language

en

Model name

Input the initial name to search for the model.

Llama-3.1-8B-Chinese-Chat

Model path

Path to pretrained model or model identifier from Hugging Face.

shenzhi-wang/Llama3.1-8B-Chinese-Chat

Finetuning method

lora

Checkpoint path

Quantization bit

Enable quantization (QLoRA).

none

Quantization method

Quantization algorithm to use.

bnb

Chat template

The chat template used in constructing prompts.

llama3

RoPE scaling

RoPE scaling method to use.

none

Booster

Approach used to boost training speed.

auto

Train

Evaluate & Predict

Chat

Export

Stage

The stage to perform in training.

Supervised Fine-Tune

Data dir

Path to the data directory.

data

Dataset

cicids_analyst

Preview dataset

Learning rate

Initial learning rate for AdamW.

5e-5

Epochs

Total number of training epochs to perform.

3.0

Maximum gradient norm

Norm for gradient clipping.

1.0

Max samples

Maximum samples per dataset.

3000

Compute type

Whether to use mixed precision training.

bf16

Cutoff length

Max tokens in input sequence.

512

Batch size

Number of samples processed on each GPU.

2

Gradient accumulation

Number of steps for gradient accumulation.

4

Val size

Percentage of validation set from the entire dataset.

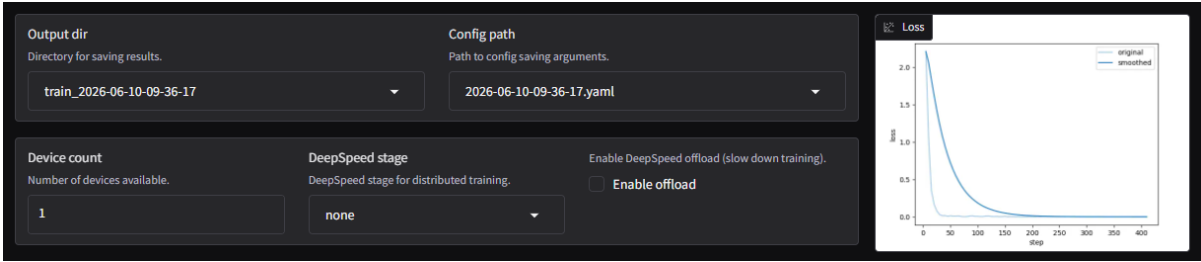
0.1

LR scheduler

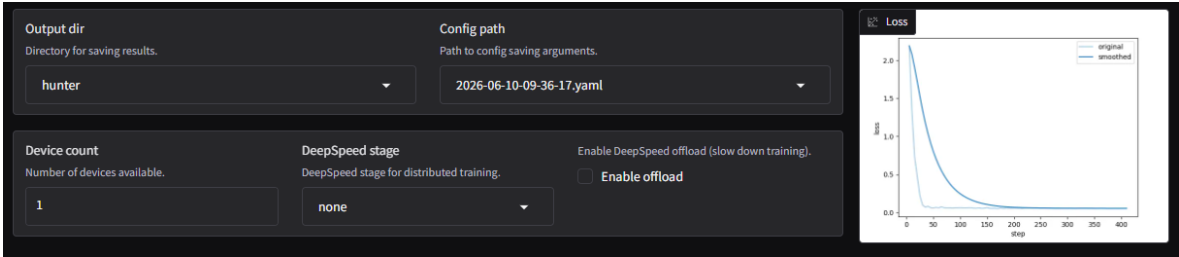
Name of the learning rate scheduler.

cosine

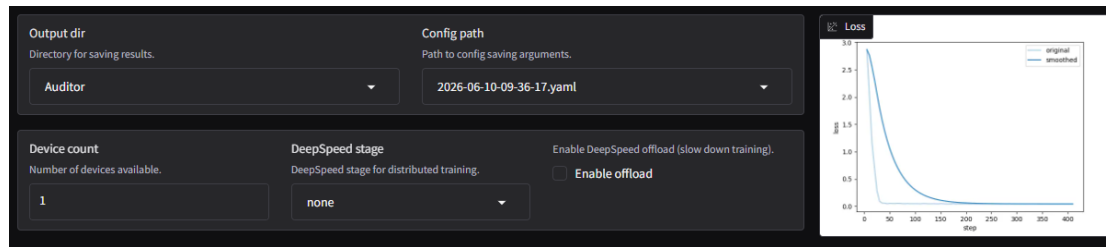
Loss diagram of Analysist dataset



Loss diagram of Hunter dataset



Loss diagram of Auditor dataset



Chat test

Test sentence

正常流量測試 測試 1 — BENIGN

Analyze the following network traffic and classify it: Flow Duration: 119261000, Total Fwd Packets: 20, Total Backward Packets: 18, Flow Bytes/s: 1024.5, Flow Packets/s: 45.2, SYN Flag Count: 1, ACK Flag Count: 35

攻擊流量測試 測試 2 — DDoS

Analyze the following network traffic and classify it: Flow Duration: 2341, Total Fwd Packets: 1500, Total Backward Packets: 0, Flow Bytes/s: 985432.1, Flow Packets/s: 6408.4, SYN Flag Count: 1500, ACK Flag Count: 0

測試 3 — PortScan

Analyze the following network traffic and classify it: Flow Duration: 51000, Total Fwd Packets: 1, Total Backward Packets: 0, Flow Bytes/s: 78.4, Flow Packets/s: 19.6, SYN Flag Count: 1, RST Flag Count: 1, Destination Port: 445

測試 4 — Brute Force (FTP)

Analyze the following network traffic and classify it: Flow Duration: 302000, Total Fwd Packets: 8, Total Backward Packets: 6, Flow Bytes/s: 231.5, Flow Packets/s: 12.3, SYN Flag Count: 1, ACK Flag Count: 13, Destination Port: 21

測試 5 — Bot

Analyze the following network traffic and classify it: Flow Duration: 7200000000,

Total Fwd Packets: 3, Total Backward Packets: 2, Flow Bytes/s: 0.8, Flow
Packets/s: 0.0007, SYN Flag Count: 1, ACK Flag Count: 4, Destination Port: 6667

```
: # 正常流量測試 測試 1 - BENIGN
parliament_judge("""
Analyze the following network flow and classify it: Flow Duration: 119261000, Total Fwd Packets: 20, Total Backward Packets: 18, Flow Bytes/s: 1024.5, Flow P
""")
#攻擊流量測試 測試 2 - Web Attack - Brute Force
parliament_judge("""
Analyze the following network flow and classify it: Flow Duration: 302000, Total Fwd Packets: 8, Total Backward Packets: 6, Flow Bytes/s: 231.5, Flow Packets:
""")
#測試 3 - Web Attack - XSS
parliament_judge("""
Analyze the following network flow and classify it: Flow Duration: 85000, Total Fwd Packets: 6, Total Backward Packets: 4, Flow Bytes/s: 980.0, Flow Packets:
""")
#測試 4 - Web Attack - Sql Injection
parliament_judge("""
Analyze the following network flow and classify it: Flow Duration: 120000, Total Fwd Packets: 5, Total Backward Packets: 3, Flow Bytes/s: 640.0, Flow Packets:
""")
#測試 5 - Bot
parliament_judge("""
Analyze the following network traffic and classify it: Flow Duration: 7200000000, Total Fwd Packets: 3, Total Backward Packets: 2, Flow Bytes/s: 0.8, Flow Packets:
""")
```

result

Test 1

```
=====
🏛️ AI Parliament Final Verdict
=====
Agreed: The traffic pattern is consistent with legitimate usage.
Challenged: Ensure no low-and-slow attack patterns are masked by normal averages. ACK/SYN ratio could match port scan or reconnaissance.
Alternative hypothesis: Possible legitimate high-throughput service or burst rather than consistent baseline.
Adjusted confidence: 0.85
Recommendation: Monitor the same source IP over a longer time window to confirm consistent behavior.
```

Test 2

```
=====
🏛️ AI Parliament Final Verdict
=====
Agreed: The traffic is consistent with a Web Attack 💎 XSS classification.
Challenged: The confidence score may be inflated. Consider whether the observed features could also match a legitimate high-throughput service.
Alternative hypothesis: Possible misconfigured client causing burst traffic rather than deliberate attack.
Adjusted confidence: 0.75
Recommendation: Correlate with endpoint logs to confirm malicious intent before escalating to incident response.
```

Test 3

```
=====
🏛️ AI Parliament Final Verdict
=====
Agreed: The traffic is consistent with a Web Attack 💎 Sql Injection attack.
Challenged: The confidence score may be inflated. Consider whether the observed features could also match a legitimate high-throughput service.
Alternative hypothesis: Possible misconfigured client causing burst traffic rather than deliberate attack.
Adjusted confidence: 0.75
Recommendation: Correlate with endpoint logs to confirm malicious intent before escalating to incident response.
```

Test 4

```
=====
🏛️ AI Parliament Final Verdict
=====
Agreed: The traffic pattern is consistent with legitimate usage.
Challenged: Ensure no low-and-slow attack patterns are masked by normal averages. ACK/SYN ratio could match port-scan or reconnaissance.
Alternative hypothesis: Possible legitimate high-throughput service or burst rather than consistent baseline.
Adjusted confidence: 0.85
Recommendation: Monitor the same source IP over a longer time window to confirm consistent behavior.
```

Test 5

```
=====
🏠 AI Parliament Final Verdict
=====
Agreed: The traffic pattern appears consistent with legitimate usage.
Challenged: Ensure no low-and-slow attack patterns are masked by normal averages. ACK/SYN ratio could match port-scan or reconnaissance.
Alternative hypothesis: Possible port-scan or reconnaissance with very low packet rate.
Adjusted confidence: 0.85
Recommendation: Monitor the same source IP over a longer time window for anomaly escalation.

'Agreed: The traffic pattern appears consistent with legitimate usage.\nChallenged: Ensure no low-and-slow attack patterns are masked by normal averages. AC
K/SYN ratio could match port-scan or reconnaissance.\nAlternative hypothesis: Possible port-scan or reconnaissance with very low packet rate.\nAdjusted conf
idence: 0.85\nRecommendation: Monitor the same source IP over a longer time window for anomaly escalation.'
```

Summary

Chat test

```
[38]: parliament_judge("", verbose=True)

Setting 'pad_token_id' to 'eos_token_id':128001 for open-end generation.
Setting 'pad_token_id' to 'eos_token_id':128001 for open-end generation.
=== Analyst ===
Summary: Network flow with 500 pkts, 300kb avg pkts size, 10min duration.
Classification: benign
Confidence: 0.90
Key indicators: packet_count, average_packet_size, flow_duration

Setting 'pad_token_id' to 'eos_token_id':128001 for open-end generation.
=== Hunter ===
ATT&CK Technique: N/A - N/A
Attack Stage: N/A
Threat Level: N/A
Hypothesis: The traffic pattern appears consistent with typical usage. No malicious indicators were identified in the available features.
Confidence: 0.9

=====
🏠 AI Parliament Final Verdict
=====
Agreed: The traffic pattern aligns with normal behavior.
Challenged: Ensure no low-and-slow attack patterns are masked by normal averages. ACK/SYN ratio should be verified across a longer time window.
Alternative hypothesis: Possible port-scan or reconnaissance with very low packet rate.
Adjusted confidence: 0.75
Recommendation: Monitor the same source IP over the next 30-minute window for anomaly escalation.

[38]: 'Agreed: The traffic pattern aligns with normal behavior.\nChallenged: Ensure no low-and-slow attack patterns are masked by normal averages. ACK/SYN ratio s
hould be verified across a longer time window.\nAlternative hypothesis: Possible port-scan or reconnaissance with very low packet rate.\nAdjusted confidenc
e: 0.75\nRecommendation: Monitor the same source IP over the next 30-minute window for anomaly escalation.'
```

測試	正確標籤	模型預測	結果
測試 1	BENIGN	BENIGN	✅ 完全正確
測試 2	DDoS	BENIGN	❌ 完全錯誤
測試 3	PortScan	Web Attack ♦ Sql Injection	❌ 大方向錯誤
測試 4	Brute Force (FTP)	Web Attack ♦ XSS	❌ 大方向錯誤
測試 5	Bot	BENIGN	❌ 完全錯誤

Score

由於 AI council 架構問題無法透過 llama factory 直接 evaluate and test，因此透過 pipeline 用 dataset 裡面 50 筆測試資料，計算六個指標。

```
[*]: from rouge_score import rouge_scorer
from nltk.translate.bleu_score import corpus_bleu, SmoothingFunction
import json, time, nltk
nltk.download('punkt_tab', quiet=True)

with open("/LLaMA-Factory/data/cicids_analyst_test.json") as f:
    test_data = json.load(f)

rouge = rouge_scorer.RougeScorer(['rouge1', 'rouge2', 'rougeL'], use_stemmer=False)
scores = {"rouge1": [], "rouge2": [], "rougeL": []}
references_bleu, hypotheses_bleu = [], []

start_time = time.time() # 開始計時

for i, item in enumerate(test_data[:50]):
    traffic = item["input"]
    reference = item["output"]

    prediction = parliament_judge(traffic, verbose=False)

    # ROUGE
    r = rouge.score(reference, prediction)
    scores["rouge1"].append(r["rouge1"].fmeasure)
    scores["rouge2"].append(r["rouge2"].fmeasure)
    scores["rougeL"].append(r["rougeL"].fmeasure)

    # BLEU
    references_bleu.append([reference.split()])
    hypotheses_bleu.append(prediction.split())

    if i % 10 == 0:
        print(f"進度: {i+1}/50")

total_time = time.time() - start_time # 總執行時間
n_samples = len(test_data[:50])

# 計算 BLEU-4
smooth = SmoothingFunction().method1
bleu4 = corpus_bleu(references_bleu, hypotheses_bleu,
                    weights=(0.25, 0.25, 0.25, 0.25),
                    smoothing_function=smooth) * 100

# 計算 BLEU-4
smooth = SmoothingFunction().method1
bleu4 = corpus_bleu(references_bleu, hypotheses_bleu,
                    weights=(0.25, 0.25, 0.25, 0.25),
                    smoothing_function=smooth) * 100

print("\n=== 🏠 AI Parliament 完整評估結果 ===")
print(f"predict_bleu-4           : {bleu4:.2f}")
print(f"predict_rouge-1           : {sum(scores['rouge1'])/n_samples*100:.2f}")
print(f"predict_rouge-2           : {sum(scores['rouge2'])/n_samples*100:.2f}")
print(f"predict_rouge-1           : {sum(scores['rougeL'])/n_samples*100:.2f}")
print(f"predict_runtime           : {total_time:.1f}s")
print(f"predict_samples_per_second : {n_samples/total_time:.2f}")
```

```
=====
🏠 AI Parliament Final Verdict
=====
Agreed: The traffic pattern is consistent with legitimate usage.
Challenged: Ensure no low-and-slow attack patterns are masked by normal averages. ACK/SYN ratio could be consistent with port scan or reconnaissance.
Alternative hypothesis: Possible legitimate high-throughput service or burst rather than consistent baseline.
Adjusted confidence: 0.85
Recommendation: Monitor the same source IP over a longer time window to confirm consistent behavior.

Setting 'pad_token_id' to 'eos_token_id':128001 for open-end generation.
Setting 'pad_token_id' to 'eos_token_id':128001 for open-end generation.
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Agreed: The traffic pattern is consistent with legitimate usage.
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Alternative hypothesis: Possible port-scan or reconnaissance with very low packet rate.
Adjusted confidence: 0.85
Recommendation: Monitor the same source IP over the next 30-minute window for anomaly escalation.

=== 🏠 AI Parliament 完整評估結果 ===
predict_bleu-4           : 0.39
predict_rouge-1           : 10.99
predict_rouge-2           : 3.40
predict_rouge-1           : 9.02
predict_runtime           : 1607.7s
predict_samples_per_second : 0.03
```

Benchmark

AI council

指標	數值	表現評級
predict_bleu-4	39	尚可
predict_rouge-1	10.99	很差
predict_rouge-2	3.40	極差
predict_rouge-l	9.02	很差
predict_runtime	1607.7s	偏慢
predict_samples_per_second	0.03	極慢

Comparison with result of baseline model

測試	正確標籤	Baseline 結果	AI council 結果
測試 1	BENIGN	✗ False Positive	✓ 完全正確
測試 2	DDoS	✓ 大方向正確	✗ 完全錯誤
測試 3	PortScan	✗ 完全錯誤	✗ 大方向錯誤
測試 4	Brute Force (FTP)	✗ 完全錯誤	✗ 大方向錯誤
測試 5	Bot	✗ 完全錯誤	✗ 完全錯誤

指標	Baseline 數值	Baseline 表現評級	AI Council 數值	AI Council 表現評級
predict_bleu-4	60.66	優異	39	優異
predict_rouge-1	64.61	優異	10.99	優異
predict_rouge-2	55.25	良好	3.40	良好
predict_rouge-l	64.54	優異	9.02	優異
predict_runtime	1049.7s	正常	1607.7s	正常
predict_samples_per_second	2.41	正常	0.03	正常

結論

透過類似 AI council 的架構，原本的預期可以使得模型正確率更高，有可能因為透過 jupyter notebook 取的資料數太少，因此導致 AI council 的架構並效能並不比 baseline 好，然而也有可能因為 AI council 的架構無法增強 AI 判斷能力。除此之外，這次 Course project 的資料集取樣有偏差，實驗控制變因沒有控制十分嚴謹，導致實驗結果可能有所偏差。